

The area of a semicircle and of a quarter circle

A fraction of the disc — and the fraction is the central angle over 360.

LESSON 100–102

3 × 40 MIN

MATEMATIKA 6, §22

S7 15

LESSON CLOCK

25'

30'

35'

25'

5

Halves and quarters	25 min
Any sector	30 min
Area against perimeter	35 min
Problems	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find the area of a semicircle and of a quadrant.
- Find the area of any sector from its central angle.
- Distinguish area from perimeter in the same figure.
- Solve problems about circular parts of real objects.

TEXTBOOK

National	pp. 291–300
Cambridge	Stage 7, pp. 148–154
Workbook	Exercise 15.3

01

Explanation

Halves and quarters

semicircle $S = \frac{1}{2}\pi r^2$ quadrant $S = \frac{1}{4}\pi r^2$

R	WHOLE DISC	SEMICIRCLE	QUADRANT
7	154	77	38.5
14	616	308	154
10	314	157	78.5

FIND THE WHOLE DISC FIRST, THEN TAKE THE FRACTION

One calculation and one division is safer than trying to remember a separate formula for each fraction.

Any sector

$$\text{sector } S = \frac{\alpha}{360} \times \pi r^2$$

ANGLE	FRACTION	AREA WITH $R = 7$
180°	$\frac{1}{2}$	77
120°	$\frac{1}{3}$	51.33
90°	$\frac{1}{4}$	38.5
60°	$\frac{1}{6}$	25.67
45°	$\frac{1}{8}$	19.25

THE SAME FRACTION AS FOR THE ARC

A 120° sector is a third of the disc's area and a third of its circumference. One fraction does both jobs.

Area against perimeter

FIGURE, $R = 7$	PERIMETER	AREA
whole disc	44 cm	154 cm ²
semicircle	36 cm	77 cm ²
quadrant	25 cm	38.5 cm ²

HALVING THE DISC HALVES THE AREA BUT NOT THE PERIMETER

The semicircle keeps half the curve and gains a diameter, so its perimeter is 36, not 22. Area is halved cleanly; boundary is not.

Problems

PROBLEM	WORKING	ANSWER
a semicircular window of radius 70 cm: the glass	$\frac{1}{2} \cdot \frac{22}{7} \cdot 4900$	7700 cm^2
a quadrant flower bed of radius 14 m	$\frac{1}{4} \cdot 616$	154 m^2
a 60° slice of a pizza of radius 21 cm	$\frac{1}{6} \cdot 1386$	231 cm^2
a fan sweeping 120° with a blade of 35 cm	$\frac{1}{3} \cdot 3850$	1283.33 cm^2

READ THE QUESTION: GLASS OR FRAME?

Glass is an area, framing is a perimeter. The same window gives two different answers depending on which is asked for.

02 Worked examples

EXAMPLE 1 Find the area of a semicircle of radius 14 cm.

1 The whole disc: $\frac{22}{7} \cdot 196 = 616\text{ cm}^2$.

2 Half of it.

3 308 cm^2

ANSWER

308 cm^2

EXAMPLE 2 Find the area of a 60° sector of a circle of radius 21 cm .

① Whole disc: $\frac{22}{7} \cdot 441 = 1386\text{ cm}^2$.

② The fraction is $\frac{60}{360} = \frac{1}{6}$.

③ $1386 \div 6 = 231\text{ cm}^2$.

ANSWER 231 cm^2 **EXAMPLE 3** A semicircular window has radius 70 cm . Find the area of glass and the length of framing.

① Glass: $\frac{1}{2} \cdot \frac{22}{7} \cdot 4900 = 7700\text{ cm}^2$.

An area.

② Framing: $\pi r + 2r = 220 + 140$.

A perimeter.

③ 360 cm of framing.

ANSWER 7700 cm^2 and 360 cm **03 Interactive model**

Ask for the glass and the framing of the same window in one question; the class learns to read for area or perimeter rather than for numbers.

Which fraction, and area or perimeter?

A semicircle of radius 7 has area:

154

77

38.5

22

A quadrant of radius 7 has area:

154

77

38.5

11

A 120° sector is what fraction?

$\frac{1}{2}$

$\frac{1}{3}$

$\frac{1}{4}$

$\frac{1}{6}$

A 60° sector of radius 21 has area:

231

462

1386

33

A semicircle of radius 7 has perimeter:

22

36

44

77

Halving a disc halves:

the perimeter

the area

both

neither

A 90° sector of radius 14 has area:

77

154

308

616

Glass in a window is measured as:

a perimeter

an area

a length

an angle

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Semicircle	Yarim doira	Полукруг
Quadrant	Chorak doira	Четверть круга
Sector	Sektor	Сектор
Central angle	Markaziy burchak	Центральный угол
Fraction of the disc	Doiraning qismi	Часть круга
Area	Yuza	Площадь
Perimeter	Perimetr	Периметр
Square units	Kvadrat birlik	Квадратные единицы

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area of a semicircle is:

$$\pi r^2$$

$$\frac{1}{2} \pi r^2$$

$$\pi r$$

$$2\pi r$$

The area of a quadrant is:

$$\frac{1}{2} \pi r^2$$

$$\frac{1}{4} \pi r^2$$

$$\frac{1}{4} \pi r$$

$$\pi r^2$$

A sector of angle α has area:

$$\frac{\alpha}{360} \pi r^2$$

$$\frac{\alpha}{180} \pi r^2$$

$$\alpha \pi r^2$$

$$\frac{360}{\alpha} \pi r^2$$

A semicircle of radius 14 has area:

$$154$$

$$308$$

$$616$$

$$88$$

A 45° sector of radius 7 has area:

$$19.25$$

$$38.5$$

$$77$$

$$154$$

Perimeter and area of the same figure are:

the same

different measurements

proportional

both lengths

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A semicircle of radius 7: the area

ANSWER 77 cm^2

2. A quadrant of radius 7

ANSWER 38.5 cm^2

3. A semicircle of radius 14

ANSWER 308 cm^2

4. A quadrant of radius 14

ANSWER 154 cm^2

5. A semicircle of radius 10 ($\pi = 3.14$)

ANSWER 157 cm^2

6. A 90° sector as a fraction

ANSWER $\frac{1}{4}$

7. A 60° sector as a fraction

ANSWER $\frac{1}{6}$

Medium · the standard question

7 PROBLEMS

1. A 60° sector of radius 21

ANSWER 231 cm^2

2. A 120° sector of radius 7

ANSWER 51.33 cm^2

3. A 45° sector of radius 7

ANSWER 19.25 cm^2

4. A semicircular window of radius 70 cm: the glass

ANSWER 7700 cm^2

5. And the framing

ANSWER 360 cm

6. A quadrant flower bed of radius 14 m

ANSWER 154 m^2

7. A 120° fan sweep with a 35 cm blade

ANSWER 1283.33 cm^2

Hard · stretch and reason

7 PROBLEMS

1. A 30° sector of radius 12 ($\pi = 3.14$)

ANSWER 37.68 cm^2

2. A sector of area 77 in a circle of radius 7: its angle

ANSWER 180°

3. A sector of area 38.5 in the same circle

ANSWER 90°

4. Half a ring between radii 7 and 14

ANSWER 231 cm^2

5. A quadrant of radius 21: area and perimeter

ANSWER 346.5 cm^2 and 75 cm

6. Why does a semicircle keep half the area but more than half the boundary?

ANSWER The diameter is added to the boundary

7. A 270° sector of radius 14

ANSWER 462 cm^2

07 Homework

Homework — 5 tasks

Find the whole disc first, then take the fraction.

- Find the area of a semicircle of radius 21 cm.
- Find the area of a quadrant of radius 28 cm.
- Find the area of a 120° sector of a circle of radius 14 cm.
- A semicircular table top has radius 60 cm. Find its area and the length of edging.
- Find the area of a 30° sector of a circle of radius 6 cm, taking $\pi = 3.14$.